



BAULKHAM HILLS HIGH SCHOOL

DECEMBER 2013
YEAR 12 TASK 1

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in questions 1 to 5
- Marks may be deducted for careless or badly arranged work

Total marks – 35

Exam consists of 3 pages.

Questions 1-5

- Attempt Questions 1-5

Topics Tested: Complex numbers

Question 1 (8 marks) - Start a new page

a) Let $z = 5 - i$ and $w = 2 + 3i$

Find in the form $a + ib$

(i) $\bar{z}w$ 2

(ii) $\frac{w}{z}$ 2

b) (i) Find the two square roots of $12 + 16i$ 2

(ii) Hence solve $z^2 - 4z + 1 - 4i = 0$ 2

Question 2 (6 marks) - Start a new page

a) (i) Express $-1 - \sqrt{3}i$ in mod-arg form 2

(ii) Hence or otherwise evaluate $(-1 - \sqrt{3}i)^8$ 2

Leave your answer in mod-arg form.

b) Given $1 - 2i$ is one root of the equation $x^2 + (1 + i)x + k = 0$, 2
find the other root and the value of k

Question 3 (8 marks) - Start a new page

a) (i) If $|z| = 1$, show that $z^n - z^{-n} = 2i \sin n\theta$ 2

(ii) Hence show that 2

$$\sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$$

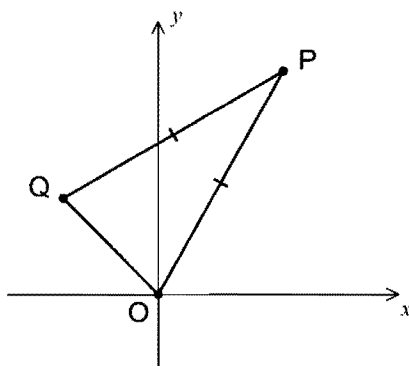
[Note: $(a - b)^5 = a^5 - 5a^4b + 10a^3b^2 - 10a^2b^3 + 5ab^4 - b^5$]

b) (i) Sketch the locus of $|z - 3i| = |z + 2 + 5i|$ 2

(ii) Find the equation of this locus. 2

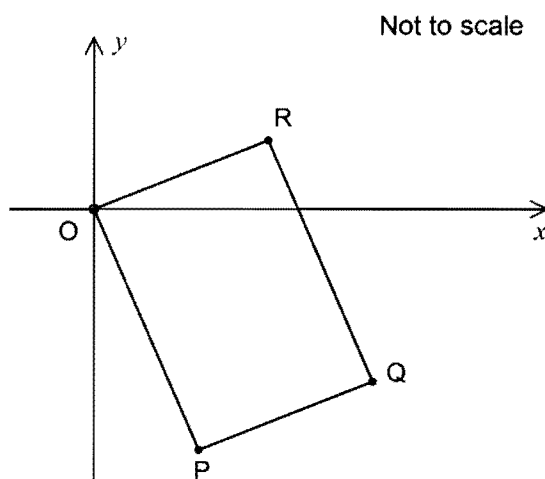
Question 4 (6 marks) - Start a new page

- a) Given $1 + \sqrt{3}i = 2\text{cis}\frac{\pi}{3}$ and $-1 - i = \sqrt{2}\text{cis}\frac{-3\pi}{4}$, find the exact value of $\cos\frac{-5\pi}{12}$ 2
- b) P is the point representing the complex number $p = 1 + \sqrt{3}i$. Triangle OPQ is isosceles with $OP = PQ$ and $\angle OPQ = \frac{\pi}{6}$ as shown in the diagram below. 2



Find the complex number representing the vector $\overrightarrow{PQ}$ in the form $a + ib$.

- c) On an Argand diagram, P represents the complex number $z = 1 - \sqrt{3}i$. 2
 $OPQR$ is a rectangle where $|OP| = 2 \times |OR|$.



Find the complex numbers representing R and Q.

Question 5 (7 marks) - Start a new page

- a) (i) Indicate on an Argand diagram the region defined by the pair of simultaneous inequalities $|z| \leq 6$ and $|z - 5| \geq 5$ 2
- (ii) Hence find the range of values of $\arg z$. 2
- b) Find the Cartesian equation of the locus of z such that $\arg(z - 2) = \arg(z^2)$. 3
 Describe the locus geometrically, noting any restrictions

- END OF PAPER -

Question 1 (8 marks)

~ Q1 - page 1 ~

BOS#:

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a) $z = 5 - i$ $w = 2 + 3i$

i) $\bar{z} \cdot w = \underbrace{(5 + i)}_{\textcircled{1}} (2 + 3i) = 10 + 3i^2 + 2i + 15i$
 $= 7 + 17i \textcircled{1}$

ii) $\frac{w}{z} = \frac{w \cdot \bar{z}}{z \cdot \bar{z}} = \frac{7 + 17i}{(5 - i)(5 + i)} \textcircled{1} = \frac{7 + 17i}{26} \textcircled{1}$

b) i) $\sqrt{12 + 16i} = z$ let $z = x + iy$

$\therefore 12 + 16i = z^2 = x^2 - y^2 + 2xyi$

(1) $12 = x^2 - y^2$

(2) $16 = 2xy \quad \therefore y = \frac{8}{x}$

$\therefore$ (1) $12 = x^2 - \frac{64}{x^2} \textcircled{1}$

$\therefore 0 = x^4 - 12x^2 - 64 \quad \therefore x^2 = \frac{12 \pm \sqrt{400}}{2} = \frac{16}{-4}$

$x^2 = 16 \quad \therefore x = \pm 4$

$x^2 = -4$ but x must be real $\therefore$ no real solns.

$\therefore x = 4 \quad \therefore y = 2 \quad \therefore z = \sqrt{12 + 16i} = 4 + 2i$

$x = -4 \quad \therefore y = -2 \quad \therefore z = \sqrt{12 + 16i} = -4 - 2i$

$\therefore \sqrt{12 + 16i} = \pm (4 + 2i) \textcircled{1}$

1b) ii) $z^2 - 4z + 1 - 4i = 0$

$\therefore z = \frac{4 \pm \sqrt{16 - 4(1 - 4i)}}{2} = \frac{4 \pm \sqrt{12 + 16i}}{2}$

$\therefore z = \frac{4 \pm (\pm 4 + 2i)}{2} \textcircled{1}$

$\therefore z = \left\{ \begin{array}{l} \frac{4 + (4 + 2i)}{2} = 4 + i \\ \frac{4 - (4 + 2i)}{2} = -i \end{array} \right\} \textcircled{1}$

You may ask for extra writing paper if you need more space to answer question 1

Question 2 (6 marks)

BOS#:

a) i) $-1 - \sqrt{3}i = 2 \operatorname{cis} \left(-\frac{2\pi}{3} \right)$
 ① ①

ii) $(-1 - \sqrt{3}i)^8 = \left[2 \operatorname{cis} \left(-\frac{2\pi}{3} \right) \right]^8 = 2^8 \operatorname{cis} \left(8 \times -\frac{2\pi}{3} \right)$

① - using De Moivre's theorem $= 2^8 \operatorname{cis} \left(-\frac{16\pi}{3} \right) = 2^8 \operatorname{cis} \left(\frac{2\pi}{3} \right)$ ① (ISE)

b) $x^2 + (1+i)x + k = 0$

let $\alpha = 1-2i$ the other root $\beta = a+ib$

$\therefore \alpha + \beta = -\frac{b}{a}$

$\therefore 1-2i + a+ib = -1-i$

$\therefore 1+a = -1$ and $-2i+ib = -i$
 $a = -2$ $b = 1$

$\therefore \beta = -2+i$ ①

now $\alpha, \beta = \frac{c}{a}$

$\therefore (1-2i)(-2+i) = k$

$-2 - 2i^2 + i + 4i = k$

$-2 + 2 + 5i = k \therefore k = 5i$ ①

Question 3 (8 marks)

BOS#:

a) i)

$|z| = 1 \therefore z = \operatorname{cis} \theta$

$\therefore z^n - z^{-n} = \operatorname{cis}(n\theta) - \operatorname{cis}(-n\theta)$ ①

$= \cos(n\theta) + i\sin(n\theta) - [\cos(-n\theta) + i\sin(-n\theta)]$

$= \cos(n\theta) + i\sin(n\theta) - [\cos(n\theta) - i\sin(n\theta)]$

$= 2i\sin(n\theta) \therefore \text{proven}$
 even ① odd

ii) since $(z - z^{-1})^5 = (2i(\sin \theta))^5$

and $(z - z^{-1})^5 = z^5 - 5z^4z^{-1} + 10z^3z^{-2} - 10z^2z^{-3} + 5z^{-4} - z^{-5}$

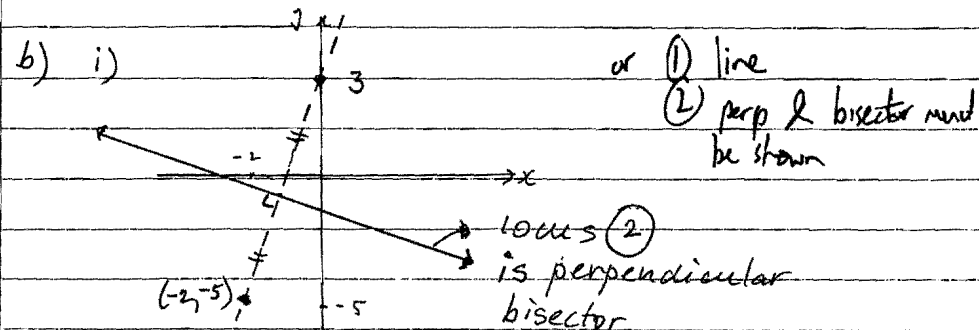
equating both ①
 $(2i\sin \theta)^5 = (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1})$

$\therefore 32i^5 \sin^5 \theta = 2i \sin 5\theta - 5 \times 2i \sin 3\theta + 10 \times 2i \sin \theta$

$\therefore \sin^5 \theta = \frac{1}{16} \sin 5\theta - \frac{10}{32} \sin 3\theta + \frac{20}{32} \sin \theta$

$\therefore \sin^5 \theta = \frac{1}{16} (\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta) \therefore \text{shown}$

b) i)



$$3b) \text{ ii) } |z-3i| = |z+2+5i| \quad \text{let } z=x+iy$$

$$|x+iy-3i| = |x+iy+2+5i|$$

$$\sqrt{x^2+(y-3)^2} = \sqrt{(x+2)^2+(y+5)^2} \quad (1)$$

$$x^2+y^2-6y+9 = x^2+4x+4+y^2+10y+25$$

$$-6y+9 = 4x+10y+29$$

$$\therefore \text{ locus is } 0 = 4x+16y+20 \quad (1)$$

$$\text{or } 0 = x+4y+5$$

which is a straight line, being perpendicular bisector of line joining $(0,3)$ & $(-2,-5)$.

Question 4 (6 marks)

BOS#:

$$a) \left. \begin{aligned} 1+\sqrt{3}i &= 2\text{cis } \frac{\pi}{3} \\ -1-i &= \sqrt{2}\text{cis } \left(-\frac{3\pi}{4}\right) \end{aligned} \right\} \text{ given}$$

$$\text{but } (1+\sqrt{3}i)(-1-i) = 2\text{cis } \frac{\pi}{3} \times \sqrt{2}\text{cis } \left(-\frac{3\pi}{4}\right)$$

$\therefore$ (1) multiplying both sides

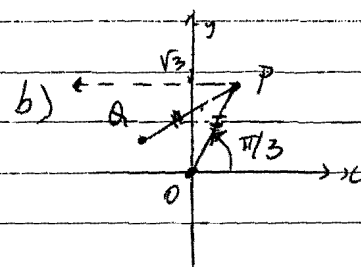
$$\therefore [-1-\sqrt{3}i+i(-\sqrt{3}-1)] = 2\sqrt{2}\left[\text{cis } \left(\frac{\pi}{3} + \left(-\frac{3\pi}{4}\right)\right)\right]$$

$$[(-1+\sqrt{3})+i(-\sqrt{3}-1)] = 2\sqrt{2}\left[\cos\left(-\frac{5\pi}{12}\right)+i\sin\left(-\frac{5\pi}{12}\right)\right]$$

now equate real parts

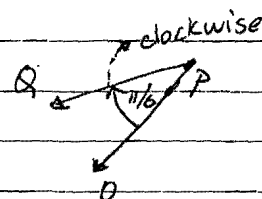
$$\therefore -1+\sqrt{3} = 2\sqrt{2}\cos\left(-\frac{5\pi}{12}\right) \quad (1)$$

$$\therefore \cos\left(-\frac{5\pi}{12}\right) = \frac{-1+\sqrt{3}}{2\sqrt{2}}$$



$$p = 1+\sqrt{3}i = \vec{OP} \quad \arg(p) = \frac{\pi}{3}$$

$$\therefore \vec{PO} = -\vec{OP} = -1-\sqrt{3}i$$



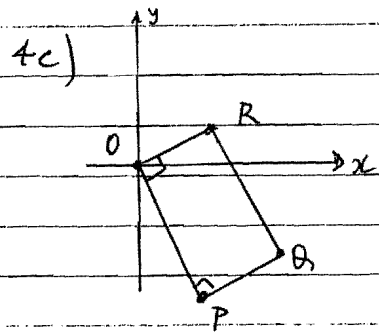
since $|OP| = |PQ|$

$$\therefore \vec{PQ} = \vec{PO} \times \text{cis}\left(-\frac{\pi}{6}\right) \quad (1)$$

$$\therefore \vec{PQ} = 2\text{cis}\left(-\frac{2\pi}{3}\right) \times \text{cis}\left(-\frac{\pi}{6}\right) = 2\text{cis}\left(-\frac{5\pi}{6}\right)$$

$$= 2\cos\left(-\frac{5\pi}{6}\right) + 2i\sin\left(-\frac{5\pi}{6}\right)$$

$$= 2 \times \left(-\frac{\sqrt{3}}{2}\right) + 2 \times \left(-\frac{1}{2}\right) = -\sqrt{3} - i \quad (1)$$



$$\vec{OP} = z = 1 - \sqrt{3}i$$

$$\therefore \vec{OR} = \frac{1}{2} \text{cis} \frac{\pi}{2} \cdot \vec{OP} = \vec{OP} \cdot \frac{1}{2}i$$

$$= \frac{1}{2}i(1 - \sqrt{3}i) = \frac{1}{2}i + \frac{\sqrt{3}}{2} = r$$

$$\therefore R\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right) \text{ is represented by } r = \frac{\sqrt{3}}{2} + \frac{1}{2}i \quad \textcircled{1}$$

$$\vec{OQ} = \vec{OP} + \vec{OR} = (1 - \sqrt{3}i) + \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$$

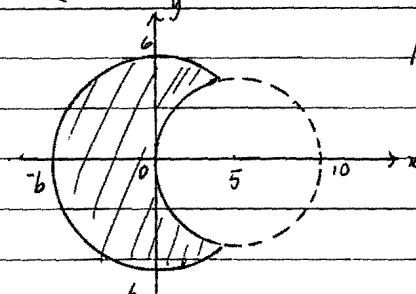
$$= \left(1 + \frac{\sqrt{3}}{2}\right) + \left(\frac{1}{2} - \sqrt{3}\right)i = q \quad \textcircled{1}$$

$$\therefore Q \text{ is represented by } q = \left(1 + \frac{\sqrt{3}}{2}\right) + \left(\frac{1}{2} - \sqrt{3}\right)i$$

Question 5 (7 marks)

BOS#:

a) i)

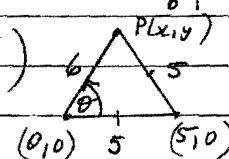


$$|z| \leq 6 \text{ and } |z-5| \geq 5$$

① - boundaries

① - shaded region

ii)



$P(x,y)$ - pt. of intersection between circles

$$\cos \theta = \frac{6^2 + 5^2 - 5^2}{2 \times 6 \times 5} = \frac{3}{5} \quad \textcircled{1} \quad \therefore \theta = \cos^{-1}\left(\frac{3}{5}\right) \approx 53.74837^\circ$$

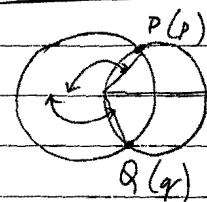
or other rounds ok

$$\therefore \cos^{-1}\left(\frac{3}{5}\right) \leq \arg z \leq \pi, \quad \textcircled{1} \quad -\pi < \arg z \leq -\cos^{-1}\left(\frac{3}{5}\right)$$

OR $53.74837^\circ \leq \arg z \leq 180^\circ, \quad -180^\circ < \arg z \leq -53.74837^\circ$

ignore if not stated cause of rounding

OR



pts of int. $P(x,y)$ $Q(x,y)$

$$\begin{cases} x^2 + y^2 = 6^2 \\ (x-5)^2 + y^2 = 25 \end{cases} \quad \begin{matrix} x = 3.6 = \frac{18}{5} \\ y = \pm 4.8 = \pm \frac{24}{5} \end{matrix}$$

$$\therefore P(3.6, 4.8) \quad \textcircled{1} \quad Q(3.6, -4.8)$$

$$\therefore p = 3.6 + 4.8i \quad q = 3.6 - 4.8i$$

$$\therefore \arg p = \tan^{-1} \frac{4.8}{3.6} \quad \arg q = -\tan^{-1} \frac{4}{3}$$

$$\arg p = \tan^{-1} \frac{4}{3}$$

$$\therefore \tan^{-1} \frac{4}{3} \leq \arg z \leq \pi, \quad -\pi < \arg z \leq -\tan^{-1} \frac{4}{3} \quad \textcircled{1}$$

OR

$$53.1301^\circ \leq \arg z \leq 180^\circ, \quad -180^\circ < \arg z \leq -53.1301^\circ$$

$$0.927... \leq \arg z \leq \pi, \quad -\pi < \arg z \leq -0.927295$$

$$5b) \arg(z-2) = \arg(z^2)$$

restriction: $z \neq 2, z \neq 0$

$\therefore$ algebraically: let $z = x+iy$ $z-2 = x-2+iy$

$$z^2 = x^2 - y^2 + 2xyi$$

$$\arg(z-2) = \tan^{-1} \frac{y}{x-2} \quad \text{and} \quad \arg z^2 = \tan^{-1} \frac{2xy}{x^2-y^2}$$

$$\therefore \tan^{-1} \frac{y}{x-2} = \tan^{-1} \frac{2xy}{x^2-y^2}$$

$$\therefore \frac{y}{x-2} \stackrel{①}{=} \frac{2xy}{x^2-y^2} \quad \left| \begin{array}{l} \text{case ① } y \neq 0 \\ \text{case ② } y = 0 \end{array} \right.$$

$$\text{case ①: } \frac{1}{x-2} = \frac{2x}{x^2-y^2}$$

$$x^2 - y^2 = 2x^2 - 4x$$

$$④ + 0 = x^2 - 4x + y^2 ④$$

since $z \neq 0$ } exclude $y \neq 0$ } $(0,0)$

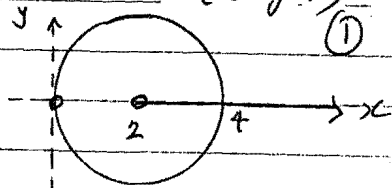
$$4 = (x-2)^2 + y^2 \quad \therefore \text{circle centre } (2,0) \quad r=2$$

case ② $y=0 \therefore x=2$, but $z \neq 2 \therefore (2,0)$ not part of locus

$$y=0 \therefore x > 2 \quad \text{is } 0 \cdot x \quad \begin{array}{c} 0 \\ 2 \end{array} \longrightarrow \text{ray}$$

$\therefore$ locus is a circle centre $(2,0)$, $r=2$

where $z \neq 0$ (excludes origin) ①. The other part of locus is a ray



purely real z
where $x > 2$

(since $z \neq 2$)

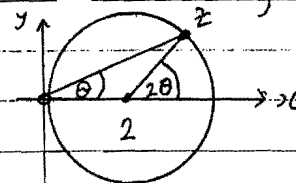
5b) geometrically

$$\text{let } z = |z| \cdot \text{cis } \theta \quad \therefore \arg z = \theta$$

$$\text{since } \arg(z^2) = 2 \times \arg z = 2 \times \theta$$

$$\therefore \text{if } \arg(z-2) = \arg z^2 = 2 \times \arg z \quad ①$$

locus of z : circle, centre $(2,0)$, radius = 2



since angle at the centre
is twice the angle
at the circumf. on same arc

$$\therefore \text{locus: } (x-2)^2 + y^2 = 4 \quad ① \text{ excluding } (0,0)$$

$$\text{① } \begin{cases} y=0 & \text{ray } x > 2 \end{cases} \longrightarrow \text{(for } z \text{ - purely real when } x > 2)$$

restrictions: $z \neq 0, z \neq 2$

OR